

MUSCLE PHYSIOLOGY

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Clinical physiology and functional investigations
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I. Introduction

Muscle tissue accounts for almost half of our body mass.

The main characteristic of muscle tissue, from a functional point of view, is its ability to convert chemical energy (in the form of ATP) into mechanical energy. Thanks to this property, muscles are able to exert a force.

Muscles can be thought of as the engines of the body.

The mobility of the body as a whole results from the activity of skeletal muscles which are distinguished from the muscles of internal organs most of which make fluids circulate in the channels of our body.

II. General Characteristics

a- Muscle Types

There are three types of muscle: skeletal, smooth and cardiac.

These three types differ in: the structure of their cells, their location in the body, their function and the way they trigger their contraction.

b- Muscle functions:

- Movement Production
- Postural maintenance
- Joint Stabilization
- Heat Release

c- Muscle functional characteristics

- **The excitability:** is the ability to perceive a stimulus and respond to it. The stimulus can be a neurotransmitter released by a nerve cell, the response is the production along the sarcolemma, of an electrical signal that is at the origin of muscle contraction.
- **The contractility:** is the ability to contract forcefully in the presence of the appropriate stimulation.
- **The extensibility:** is the faculty of stretching; When they contract, muscle fibers shorten, but when they are relaxed, they can be stretched beyond their resting length.
- **The elasticity:** is the ability of muscle fibers to regain their resting length when relaxed.

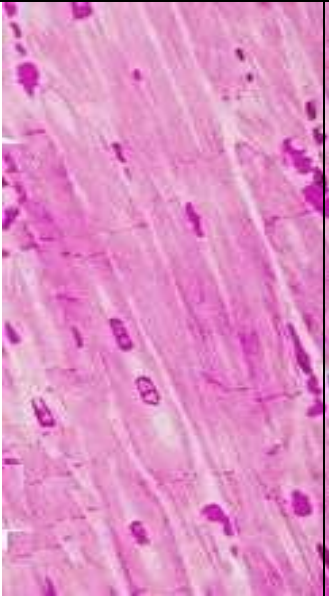
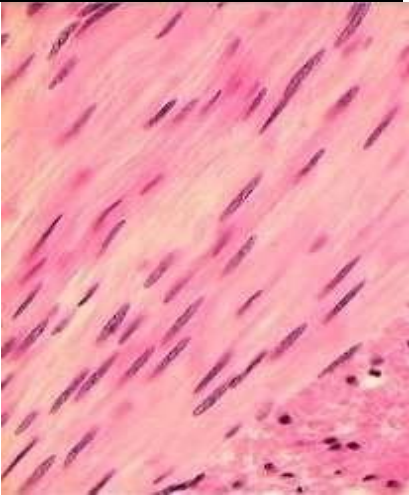
	Skeletal muscle	Cardiac muscle	Smooth muscle
Location	Pick up the Bone Skeleton and stick to it	Heart	In the walls of visceral organs (stomach, bladder, uterus), airways and blood vessels
Voluntary or involuntary?	Volunteer	Involuntary	Involuntary
Appearance			
Striped	Yes	Yes	No
Contraction	Can contract quickly but tires easily	It contracts at a relatively steady rate	Slow and continuous contractions (does not get tired)

Table : Differences between the three types of muscle tissue

III- Skeletal Muscle

A- Macroscopic Anatomy

Skeletal muscle is a well-demarcated organ, enclosing blood vessels, neurofibers and a large amount of connective tissue.

A - 1 Connective tissue envelopes:

-Each muscle fiber is inside a thin sheath of connective tissue called **endomysium**. Several fibers and their endomysium are placed side by side and form a set called **beam**.

-Each bundle is in turn bounded by a thicker sheath of connective tissue called **perimysium**. The bundles are grouped into a coarser lining composed of denser connective tissue that surrounds the entire muscle called **epimysium**.

A – 2 Innervation and vascularization

- Each muscle fiber is equipped with a nerve ending that controls its activity.
- Each muscle is served by an artery and one or more veins.
Arteries carry nutrients and oxygen, while veins remove metabolic waste

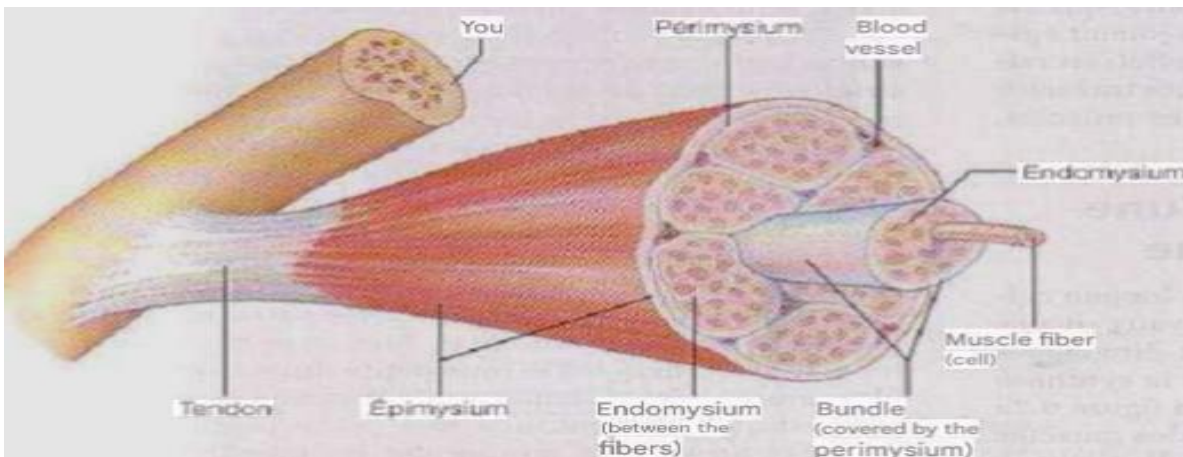


Figure 1 : Connective tissue envelopes of a muscle

B- Microscopic anatomy of a skeletal muscle fiber

B - 1 The Muscle Fiber

- Each muscle fiber is a long cylindrical cell enclosing many nuclei. It is surrounded by a membrane: the sarcolemma

-The sarcoplasm of a muscle fiber is comparable to the cytoplasm of other cells, but it contains large stores of glycogen as well as myoglobin, a protein that binds to oxygen and does not exist in any other type of cell.

B - 2 Myofibrils: Each muscle fiber contains a large number of parallel myofibrils that run the entire length of the cell.

B - 3 Myofilaments : Along the length of each myofibril, we notice a alternating dark and light bands called stripes.

- Dark bands are called bands **A (dashes A)**, the light bands named **I (dashed I)**.

Each band A has in its middle a lighter area is the light area or **striped H (zone H)**.

Each light area is divided in two by a dark line or **M lineage**.

- In the middle of the I bands, there is also a darker area called: line **Z**
- The region of a myofibril between two successive Z-lines is called **Sarcomere**, measuring 2 μm and represents the functional unit of the muscle.
- At the molecular level each myofibril is made up of filaments arranged in a very regular way: **thin and thick filaments**. Thick filaments are made up of the assembly of protein molecules: **The myosin** (present in the A bands, in the center of the sarcomere), while the essential constituent of thin filaments is another protein: **actin** ; present in bands I)

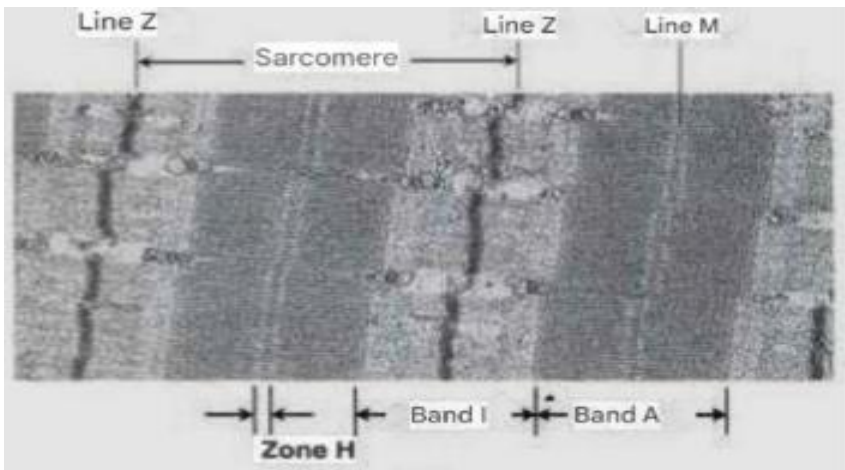


Figure 2: Electron microscope view of a myofibril

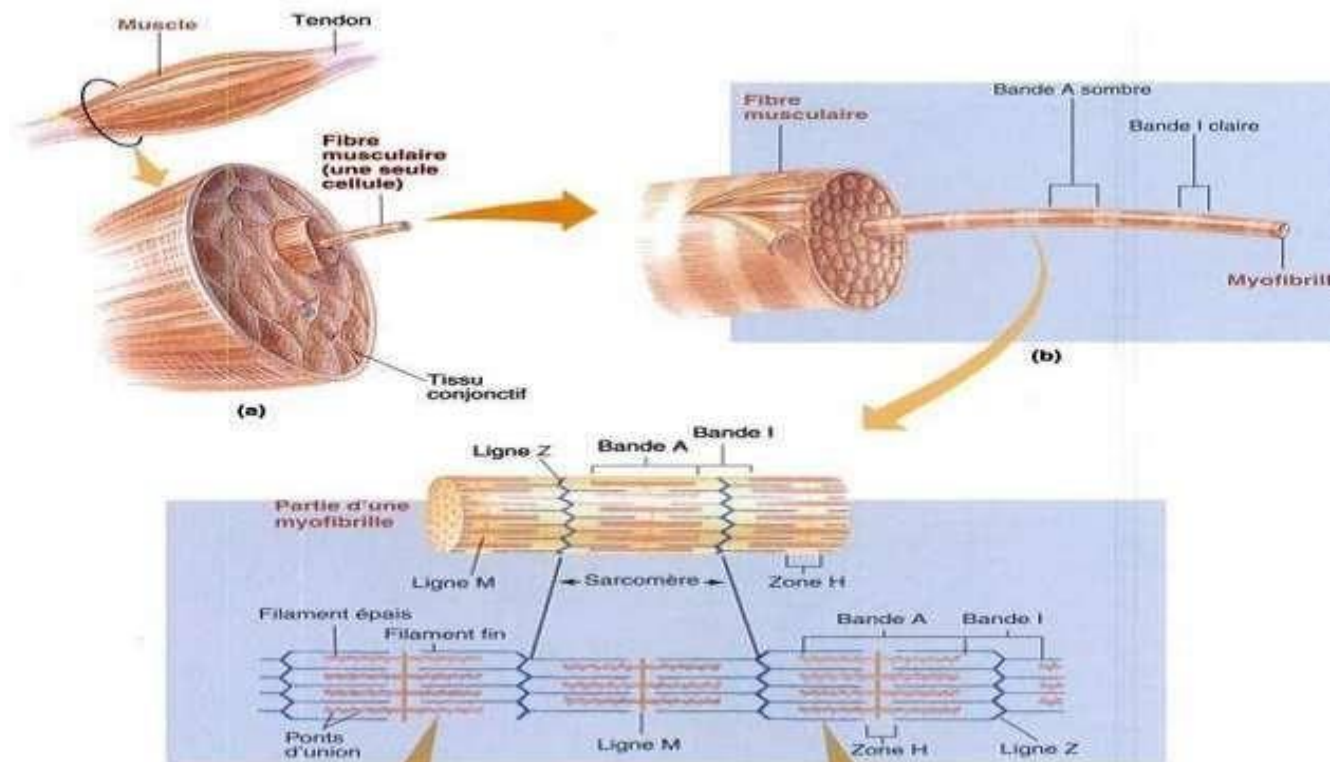


Figure 3: Levels of organization of skeletal muscle

The thick filaments:

myosin made of two identical sub-units in the shape of a golf club whose tails are wound on top of each other and whose two spherical heads protrude at one of the ends.

Myosin heads contain actin binding sites, ATP as well as ATPases enzymes that dissociate ATP to produce the energy needed for muscle contraction. Heads are also involved in cross-bridge formation.

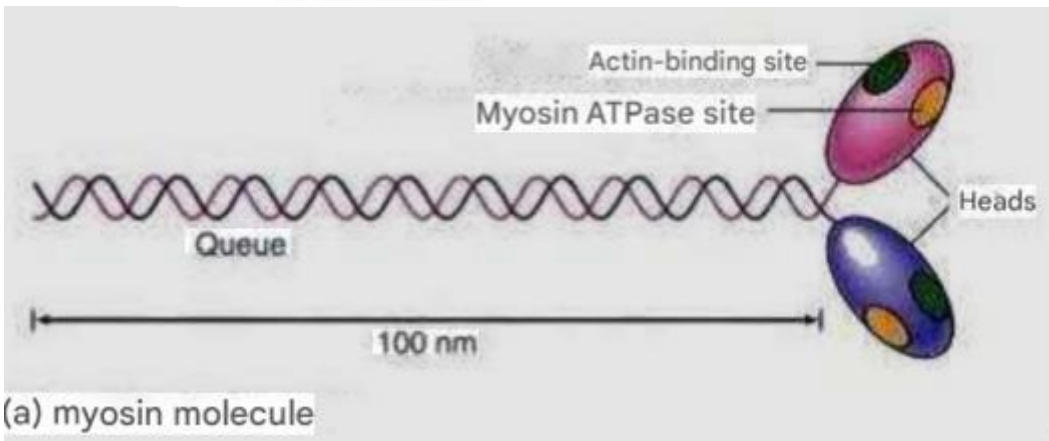
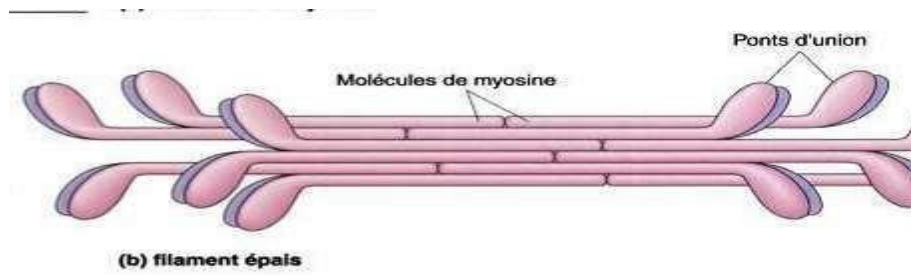


Figure 4 : thick filament (myosin)

- **Fine or thin filaments:** They are made up of three proteins: F-actin (filamentous), troponin and tropomyosin.

-Two filamentous actin chains made of a succession of globular actin molecules that are coiled around each other.

-Actin carries binding sites onto which myosin heads (bonding bridges) attach during contraction.

-Troponin molecules which consist of three spherically shaped subunits (**TnI** = inhibitory, binds to actin, **TnT** = binds to tropomyosin, **TnC** = binds to calcium ions).

-Tropomyosin filaments form a ribbon resting on the sill of the actin helix and block (mask) the actin active sites so that the myosin heads cannot bind with the thin filaments.

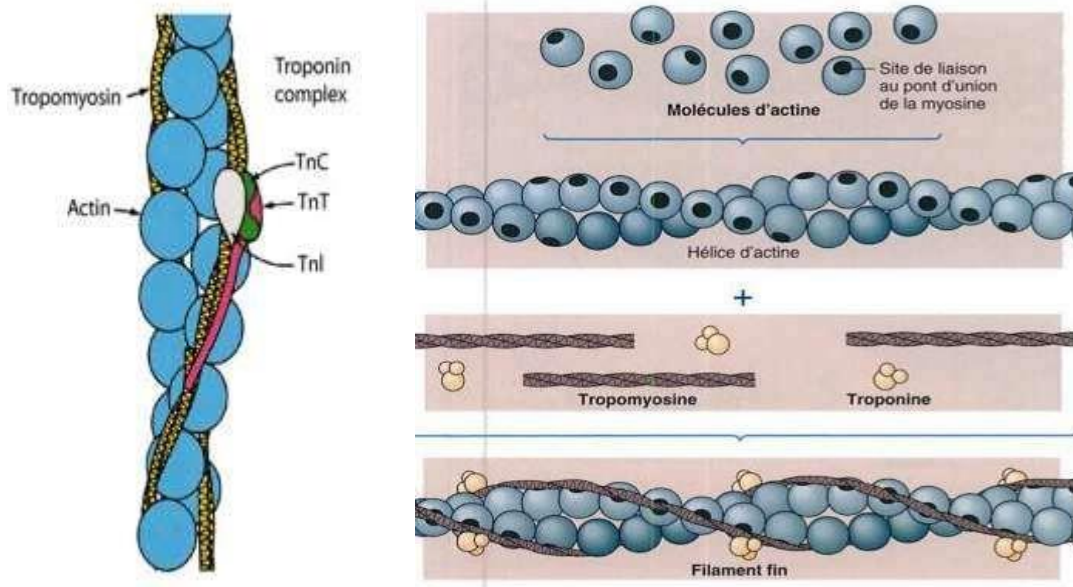


Figure 5 : filament mince (Actine)

B - 4 Sarcoplasmic reticulum and transverse T tubules:

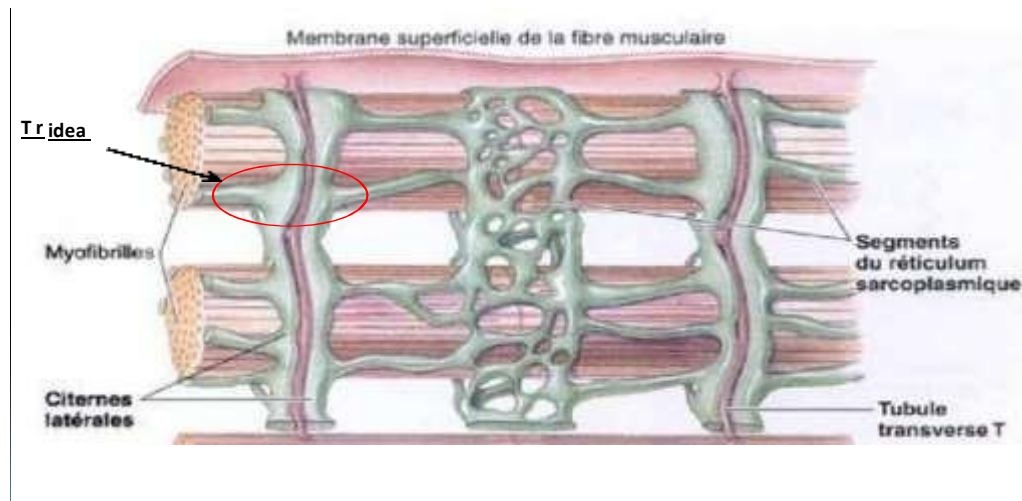
-**Transverse tubules T** : at the junction of striae A and I, the sarcolemma has invaginations that penetrate the inside of the muscle fiber and form the Transverse Tubules (T tubule).

- **The sarcoplasmic reticulum (SR)**: is a special endoplasmic reticulum, which forms a network of fine tubules surrounding each myofibril along its length. It has terminal cisternae that establish intimate contacts with the T-tubules. The T-tubule and the lateral cisternae located on each side form: **The triad**

- The sarcoplasmic reticulum plays a key role in :
- * The storage of calcium: inside the RS, calcium is bound to a protein :

The socks

- * The release of calcium from the sarcoplasm
- *The reuptake of calcium: from the intracellular environment to the inside of the RS using pumps calcium ATP- dependent.



Figs. 6 : Sarcoplasmic reticulum and T-tubules

- Role of the Triad in the transmission of information:

The transmission of information (nerve inflow) from the tubular system to the sarcoplasmic reticulum (which leads to the release of intracellular calcium) involves a specific mechanism. The following are in play:

- Voltage-dependent channels localized in the tubular membrane and blocked by dihydropyridine (DHP) hence their name DHP or dihydropyridine receptor channels (**DHPR**).
- Each dihydropyridine receptor (DHPR) is in contiguity with a Ryanodine-sensitive sarcoplasmic reticulum calcium channel hence their channel name Ryanodine or Ryanodine receptor (**RyR**)
- Under the effect of membrane depolarization, DHPR acts as an electric current or voltage intensity sensor and undergoes conformational changes leading to molecular interaction with the RyR.

This promotes its opening and the release of calcium from the stores of the sarcoplasmic reticulum and

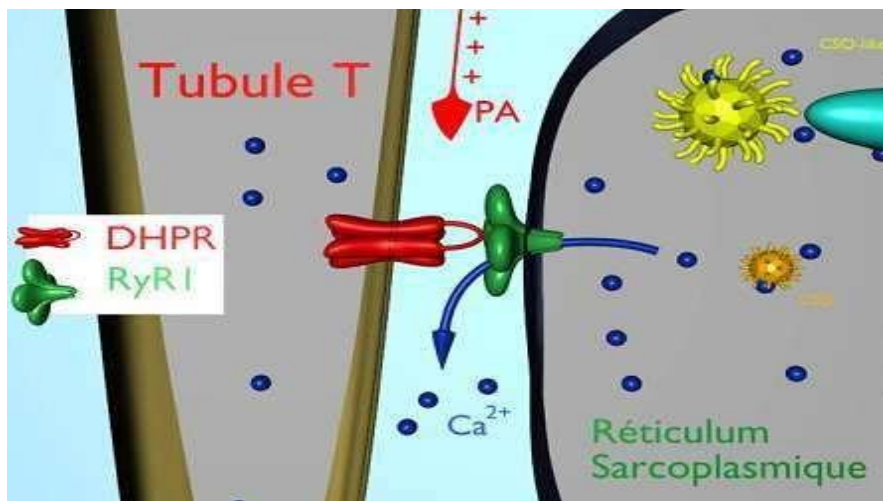


Figure 7 : Transmission of information from the tubular system to the RS

C- Mechanisms of contraction:

C -1- Filament sliding model

The theory of filament sliding contraction, developed by **Hugh Huxley in 1954**, proposes the following explanation: “during contraction, the thin filaments (actin) slide along the thick filaments (myosin), so that the actin and myosin filaments further overlap”.

At rest the thick and thin filaments overlap only a small portion of their lengths, but when muscle cells are stimulated, the myosin heads latch onto the actin binding sites and sliding begins.

The myosin heads pull the thin filaments towards the center of the sarcomere: This is the **sarcomere shortening**. The length of the A bands does not change during shortening but that of the I and H bands decreases.

C-2 Excitation- contraction coupling :

It is the sequence of events by which the action potential transmitted along the sarcolemma causes the myofilaments to slip.

The excitation - contraction coupling goes through the following steps :

- 1 The action potential propagates along the sarcolemma and transverse tubules.
- 2 When the action potential reaches the triads, and **DHPR** acts as a detector of electrical current or voltage intensity and undergoes conformational changes leading to molecular interaction with the **RyR**. This promotes its opening and the release of calcium from the stores of the sarcoplasmic reticulum and

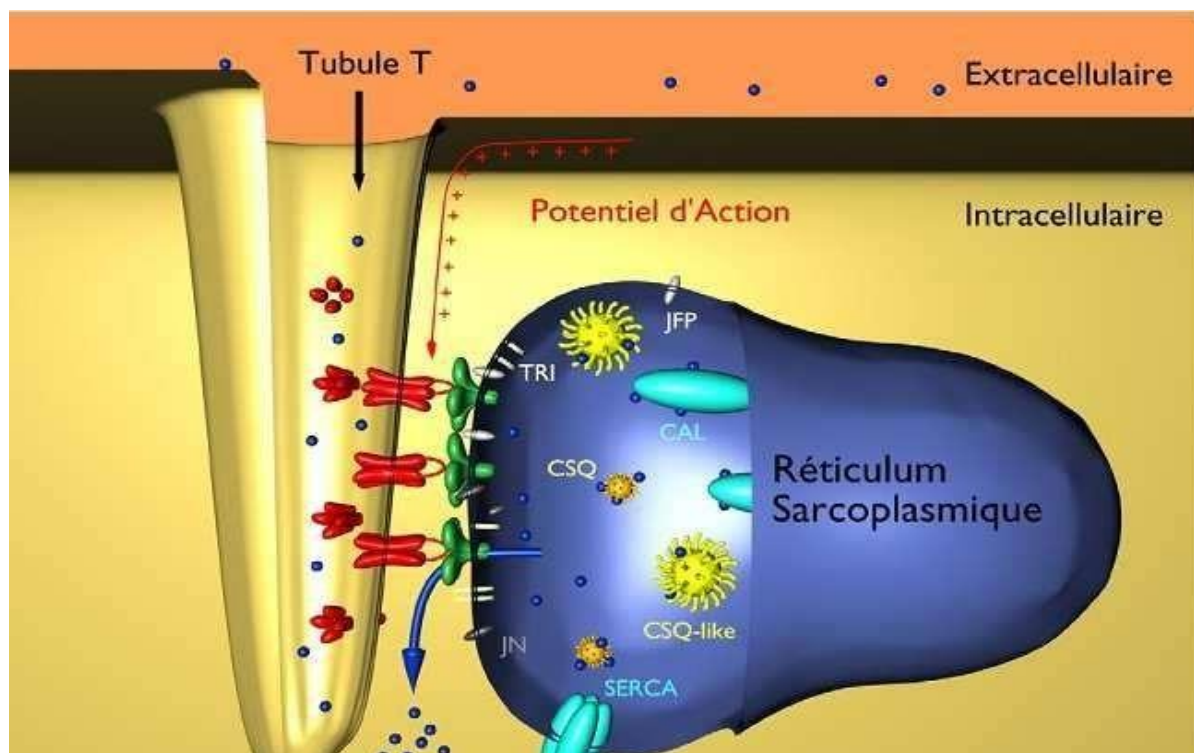


Figure 8 : Mechanism of calcium release by RS

- 3 Once in the intracellular environment (sarcoplasm), calcium binds to troponin C (TnC), four molecules of calcium bind to one molecule of TnC.
- 4 Troponin then changes its three-dimensional structure, causing the lateral displacement of tropomyosin and thus the release (unmasking) of actin binding sites.
- 5 As soon as the actin binding sites are exposed, the myosin heads se immediately attach to actin forming a complex: actomyosin
- 6 At the same time the binding of calcium on TnC allows the release of the inhibition exerted by troponin I on the ATPase activity of the myosin head, this ATPase activity allows the hydrolysis of ATP into ADP and Pi (this reaction is Mg⁺ dependent).
- 7 The detachment of Pi and ADP from the myosin heads, allows the bending of the myosin heads (changes in the angle formed by the myosin heads attached to actin) = **sliding of actin filaments on myosin filaments**)
- 8 The binding of actin to myosin remains stable and only the presence of a new molecule of ATP(which attaches to the myosin head), allows the breaking of the binding between actin and myosin.
- 9 The contraction continues as long as the calcium signal and ATP are present.
- 10 In the absence of an action potential, the sarcoplasmic reticulum recovers calcium from the sarcoplasm, troponin changes its shape again and tropomyosin masks the binding sites of the myosin heads to actin so contraction ends and the filaments resume their initial position = Relaxation of muscle

NB : Cadaveric stiffness well illustrates the fact that it is ATP that enables the detachment of myosin heads from actin. Indeed after death ATP synthesis ceases and the detachment of myosin heads becomes impossible. Actin and myosin are then irreversibly bound, resulting in cadaveric stiffness, which disappears when the muscle proteins break down a few hours after death.

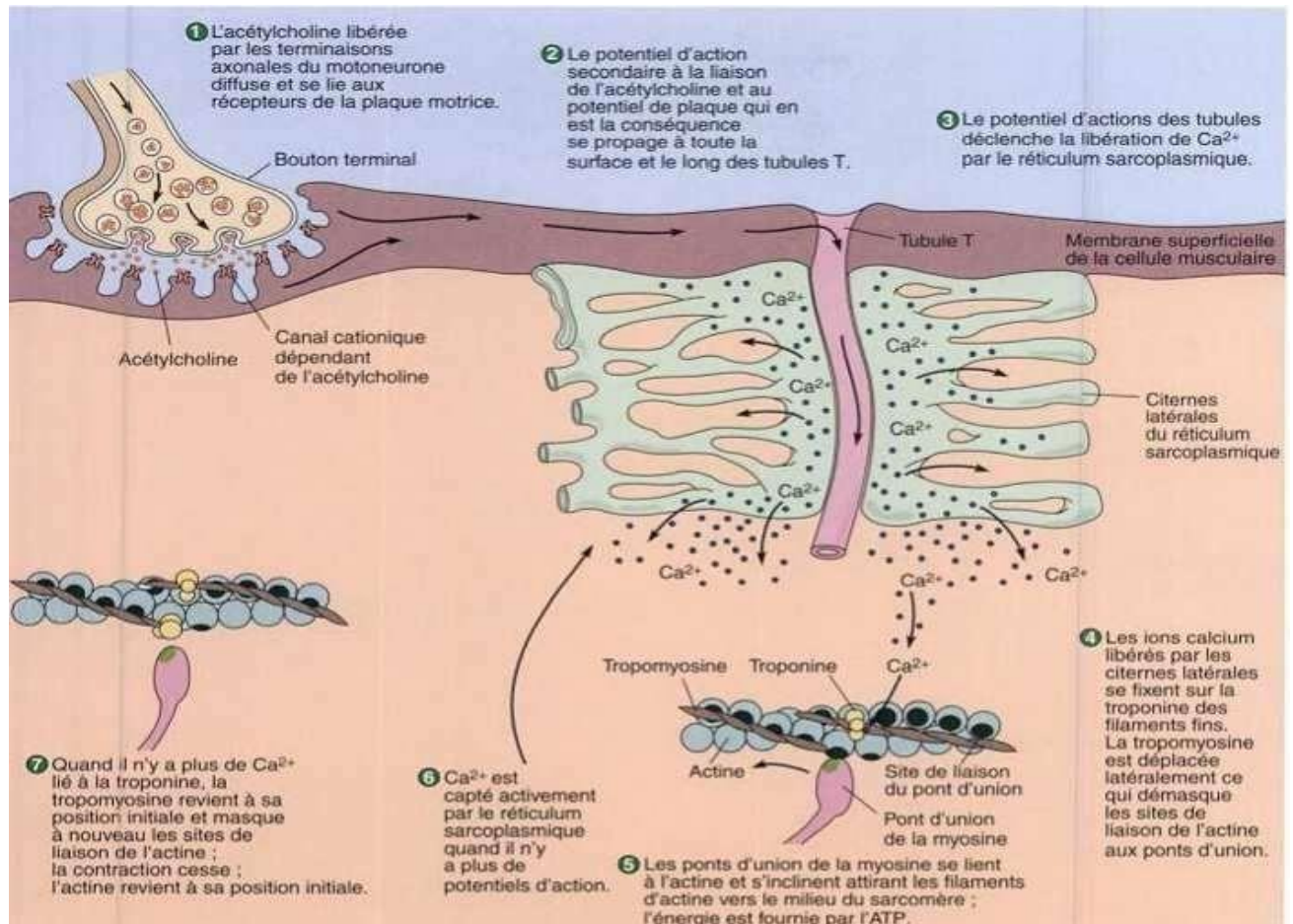


Figure 9 : excitation – contraction coupling

D- Skeletal Muscle Metabolism: ATP Regeneration

-During muscle contraction the energy used for contractile activity (flexion, detachment of myosin heads and operation of the calcium pump) is provided by ATP.

-Given that ATP is the only energy source that can directly fuel contraction, and that immediately available ATP stores are small in the muscle and allowing for a 4- to 6-second contraction, it must be regenerated continuously so that the contraction can continue. Fortunately its regeneration is done in a fraction of a second **three ways**:

a- Interaction of ADP with creatine phosphate(CP)

- At the beginning of the contraction, once the low ATP stores have been consumed, additional ATP is quickly reconstituted from a high-energy molecule: Creatine-phosphate (CP):



- This reaction is catalyzed by Creatine Kinase.
- A maximum muscle power can be maintained for 10 to 15s

b- Anaerobic Glycolysis

Muscle glycogen stores are converted into lactic acid with the production of two (02) molecules of ATP. Together, ATP and creatine phosphate stores and aerobic glycolysis can sustain muscle activity for one minute.

a- Aerobic cellular respiration: oxidative phosphorylation

During mild but prolonged muscle activity, the ATP used by the muscles is provided by aerobic cellular respiration which takes place in the mitochondria and requires the presence of oxygen and involves a series of chemical reactions (Krebs cycle, respiratory electron transport chain). During aerobic respiration glucose is entirely broken down; the complete oxidation of one molecule of glucose provides 36 molecules of ATP.

CRÉATINE PHOSPHATE	GLYCOLYSE	RESPIRATION
Source d'énergie: CP	Source d'énergie: glucose	Source d'énergie: glucose; acide pyruvique; acides gras libres provenant du tissu adipeux; acides aminés provenant du catabolisme protéinique
Utilisation d'oxygène: aucune Produits: 1 ATP par CP, créatine Durée de la réserve d'énergie: 15 s	Utilisation d'oxygène: aucune Produits: 2 ATP par glucose; acide lactique Durée de la réserve d'énergie: 30 à 60 s	Utilisation d'oxygène: nécessaire Produits: 36 ATP par glucose, CO ₂ , H ₂ O Durée de la réserve d'énergie: plusieurs heures

Table : ATP regeneration pathways

E-Types of muscle fibers:

Slow-twitch oxidative fibers (type I)

- Red color, - Plenty of myoglobin.
- Low glycogen stores - Great fatigue resistance.
- The aerobic pathway is the main pathway of ATP synthesis

Fast-twitch oxidative fibers (type IIa)

- Red color - Significant glycogen stores.
- Abundant myoglobin - Moderate fatigue resistance.

Fast-twitch glycolytic fibers (type IIb)

- Significant glycogen stores. - Low myoglobin content
- White Color - Fatiguing

II-Smooth Muscle Tissue

a- Structure of smooth muscle

-Present in the wall of most hollow organs of the body: Airways, vessels, digestive tract, and genito-urinary tract.

-Each smooth muscle fiber is a spindle cell containing a single nucleus, the diameter of these small fibers is between 2 and 4 μm .

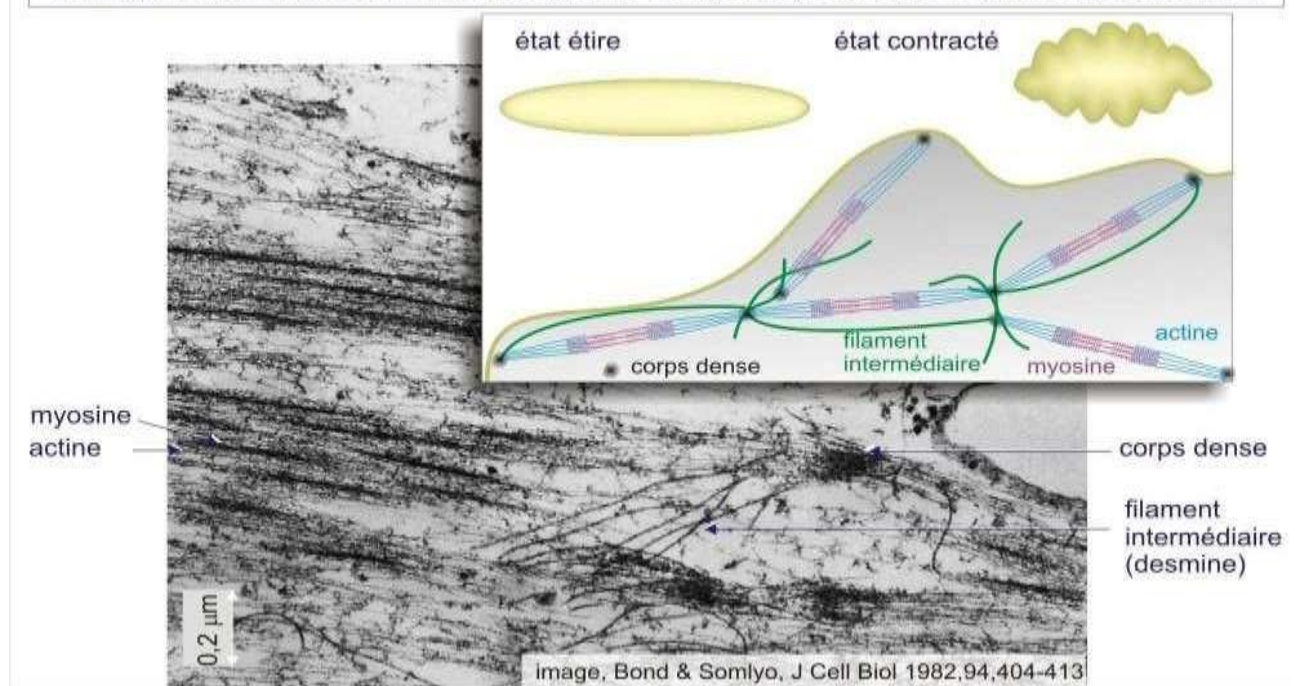
-Smooth muscle fibers do not have highly elaborate nerve endings such as those found in skeletal muscles, however they are connected to neurofibers of the autonomic nervous system.

-The sarcoplasmic reticulum of smooth muscle fibers is less developed than that of skeletal muscle fibers so there are no transverse T-tubules.

-smooth muscles do not have transverse striae, some contain thick and thin filaments but these filaments are different from those found in skeletal muscles:

- Thick and thin filaments are not arranged in sarcomeres
- The thick filaments of smooth muscle carry myosin heads along their entire length, a feature that allows these muscles to be so powerful.
- Tropomyosin is associated with thin filaments but not troponin.

arrangement des unités contractiles (actine et myosine) dans une cellule musculaire lisse



Structure of smooth muscle

b- Excitation-contraction coupling in smooth muscle:

The mechanism of smooth muscle contraction is similar to that of muscle skeletons, and this on the following planes:

- Myofilament sliding is due to the interaction of actin and myosin
- Contraction is triggered by increasing intracellular concentration of calcium ions
- Myofilament sliding requires ATP

- **The steps of muscle contraction:**

- During excitation-contraction coupling, calcium is released from the sarcoplasmic reticulum, but it also penetrates from the interstitial fluid
- To activate myosin, calcium interacts with regulatory proteins : **the calmodulin** located on myosin filaments and a kinase called **Myosin light chain kinase (MLCK)**,
- And since the thin filaments have no troponin to mask the binding site of the myosin heads and are therefore always ready to contract

- Calcium binds to calmodulin and the calcium-calmodulin complex binds to and activates myosin light chain kinase (MLCK).
- The activated kinase hydrolyzes ATP and phosphorylates myosin which allows it to interact with actin: shortening occurs.
- Like skeletal muscle, smooth muscle relaxes when the intracellular calcium concentration decreases.

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